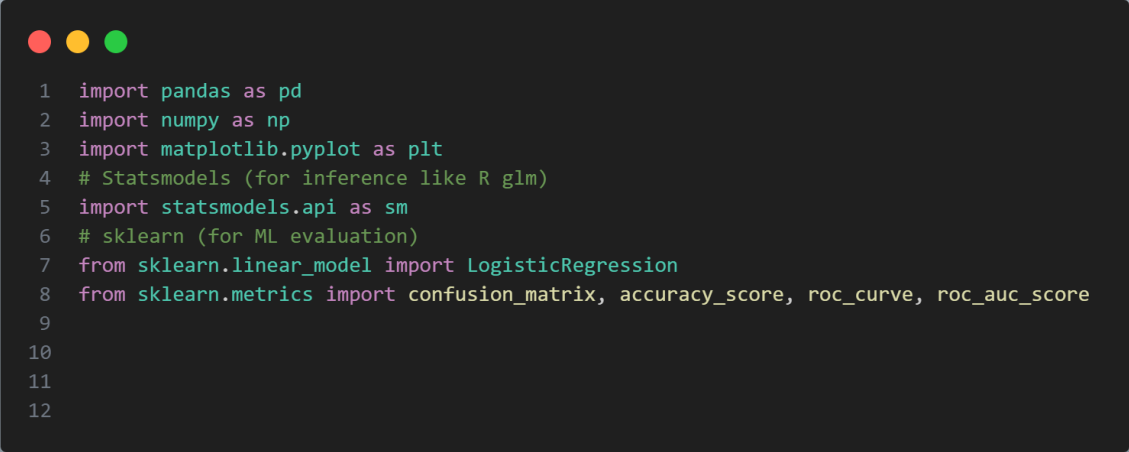


Name :- Ammar

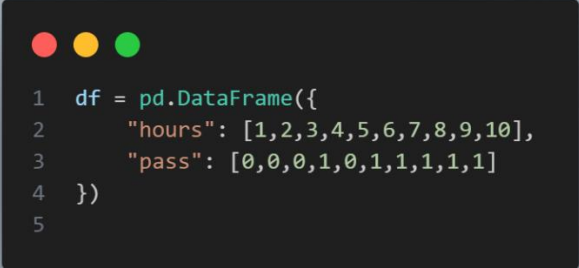
Course :- MLT (Artificial Intelligence)

Libraries :-

A code editor window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. It contains Python code for importing various libraries.

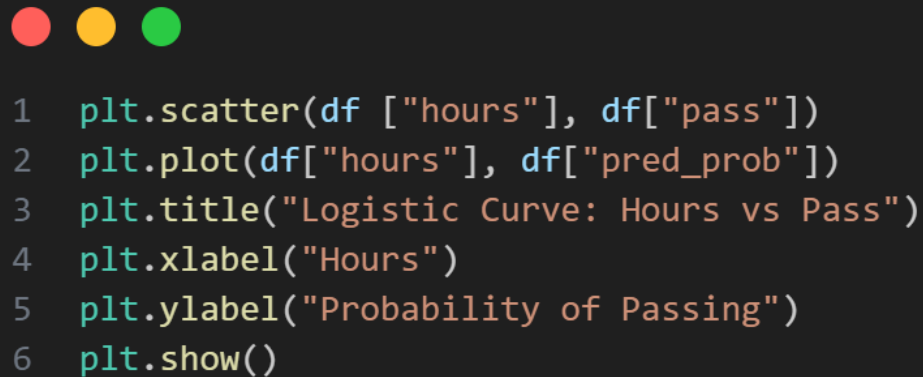
```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 # Statsmodels (for inference like R glm)
5 import statsmodels.api as sm
6 # sklearn (for ML evaluation)
7 from sklearn.linear_model import LogisticRegression
8 from sklearn.metrics import confusion_matrix, accuracy_score, roc_curve, roc_auc_score
9
10
11
12
```

Sample DataFrame :-

A code editor window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. It contains Python code for creating a pandas DataFrame.

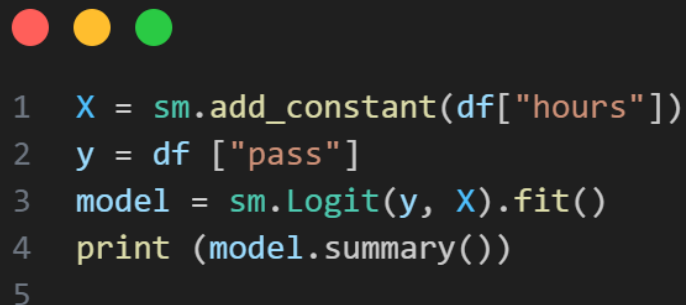
```
1 df = pd.DataFrame({
2     "hours": [1,2,3,4,5,6,7,8,9,10],
3     "pass": [0,0,0,1,0,1,1,1,1,1]
4 })
5
```

Visualization (Logistic Curve) :-



```
1 plt.scatter(df ["hours"], df["pass"])
2 plt.plot(df["hours"], df["pred_prob"])
3 plt.title("Logistic Curve: Hours vs Pass")
4 plt.xlabel("Hours")
5 plt.ylabel("Probability of Passing")
6 plt.show()
```

STATISTICAL Model (GLM like R) :-



```
1 X = sm.add_constant(df["hours"])
2 y = df ["pass"]
3 model = sm.Logit(y, X).fit()
4 print (model.summary())
5
```

Odds Ratio :-



```
1 odds_ratio= np.exp(model.params)
2 print("\nOdds Ratio:\n", odds_ratio)
```

Machine Learning Model :-



```
1 clf = LogisticRegression()
2 clf.fit(df[["hours"]], y)
```

Predictions :-

```
1 df["pred_prob"] = clf.predict_proba(df[["hours"]])[:,1]
2 df["pred_class"] = clf.predict(df[["hours"]])
```

Evaluation :-

```
1 print("\nAccuracy:", accuracy_score(y, df["pred_class"]))
2 print("\nConfusion Matrix:\n", confusion_matrix(y, df["pred_class"]))
3
```

ROC Curve :-

```
1 fpr, tpr, _ = roc_curve(y, df["pred_prob"])
2 auc = roc_auc_score(y, df["pred_prob"])
3
4 plt.plot(fpr, tpr)
5 plt.title(f"ROC Curve (AUC = {auc:.2f})")
6 plt.xlabel("False Positive Rate")
7 plt.ylabel("True Positive Rate")
8 plt.show()
9
```

2. Attendance vs Pass :-

Optimization terminated successfully.

Current function value: 0.250690

Iterations 8

Logit Regression Results

=====			
Dep. Variable:	pass	No. Observations:	10
Model:	Logit	Df Residuals:	8
Method:	MLE	Df Model:	1
Date:	Wed, 08 Apr 2026	Pseudo R-squ.:	0.6275
Time:	13:36:12	Log-Likelihood:	-2.5069
converged:	True	LL-Null:	-6.7301
Covariance Type:	nonrobust	LLR p-value:	0.003658
=====			

	coef	std err	z	P> z	[0.025	0.975]

const	-17.4835	11.482	-1.523	0.128	-39.988	5.021
attendance	0.2591	0.169	1.533	0.125	-0.072	0.590
=====						

Odds Ratio:

const 2.552686e-08

attendance 1.295747e+00

dtype: float64

<matplotlib.collections.PathCollection object at 0x000001C4438B79D0>

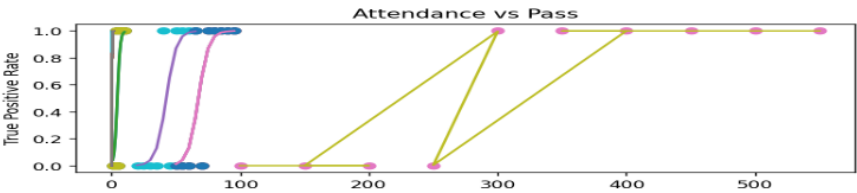
Text(0.5, 1.0, 'Attendance vs Pass')

Accuracy: 0.8

[[3 1]

[1 5]]

[<matplotlib.lines.Line2D object at 0x000001C43E44EC10>]



Logit Regression Results

Dep. Variable:	purchase	No. Observations:	10
Model:	Logit	Df Residuals:	8
Method:	MLE	Df Model:	1
Date:	Wed, 08 Apr 2026	Pseudo R-squ.:	1.000
Time:	14:03:45	Log-Likelihood:	-1.1706e-09
converged:	False	LL-Null:	-6.7301
Covariance Type:	nonrobust	LLR p-value:	0.0002437

coef	std err	z	P> z	[0.025	0.975]
------	---------	---	------	--------	--------

const	-233.8024	3.22e+05	-0.001	0.999	-6.3e+05	6.3e+05
spend	0.8505	1173.976	0.001	0.999	-2300.100	2301.801

Odds Ratio:

```
const 2.890022e-102
```

spend	2.340869e+00
-------	--------------

dtype: float64

LogisticRegression()

```
<matplotlib.collections.PathCollection object at 0x000001C444BC87D0>
```

```
[<matplotlib.lines.Line2D object at 0x000001C444BC8A50>]
```

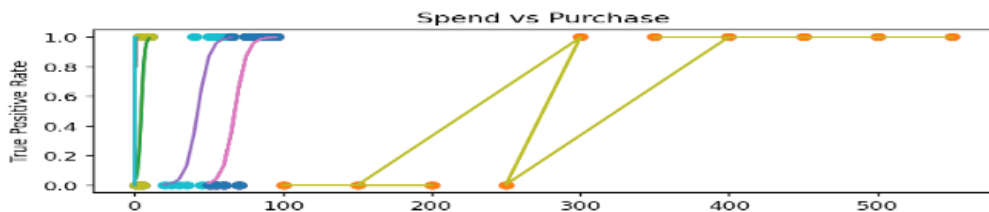
Text(0.5, 1.0, 'Spend vs Purchase')

Accuracy: 1.0

[[4 0]]

[0 6]]

```
[<matplotlib.lines.Line2D object at 0x000001C43E44F890>]
```



4 Website Time vs Signup :-

Optimization terminated successfully.

Current function value: 0.250690

Iterations 8 Logit Regression Results

=====

Dep. Variable:	signup	No. Observations:	10
Model:	Logit	Df Residuals:	8
Method:	MLE	Df Model:	1
Date:	Wed, 08 Apr 2026	Pseudo R-squ.:	0.6275
Time:	14:08:38	Log-Likelihood:	-2.5069
converged:	True	LL-Null:	-6.7301
Covariance Type:	nonrobust	LLR p-value:	0.003658

=====

	coef	std err	z	P> z	[0.025	0.975]
const	-5.8246	3.986	-1.461	0.144	-13.637	1.988
time	1.2954	0.845	1.533	0.125	-0.361	2.952

=====

Odds Ratio:

const 0.002954

time 3.652592

dtype: float64

LogisticRegression()

<matplotlib.collections.PathCollection object at 0x000001C444BC8F50>

[<matplotlib.lines.Line2D object at 0x000001C444BC9090>]

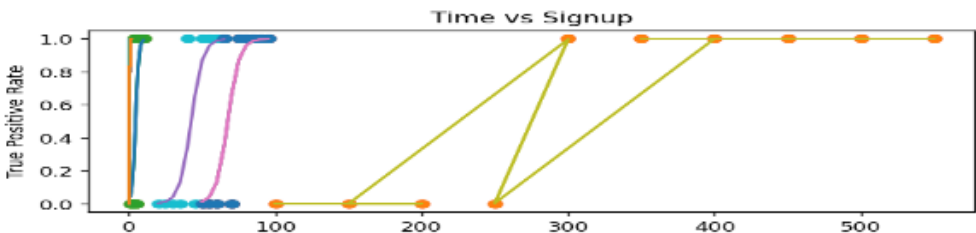
Text(0.5, 1.0, 'Time vs Signup')

Accuracy: 0.8

[[3 1]

[1 5]]

[<matplotlib.lines.Line2D object at 0x000001C444BC91D0>]



5 Age vs Disease :-

Optimization terminated successfully.

Current function value: 0.250901

Iterations 8 Logit Regression Results

```
=====
Dep. Variable:      disease  No. Observations:      10
Model:              Logit  Df Residuals:           8
Method:             MLE    Df Model:              1
Date:              Wed, 08 Apr 2026  Pseudo R-squ.:    0.6380
Time:              14:13:13  Log-Likelihood:      -2.5090
converged:          True  LL-Null:                -6.9315
Covariance Type:    nonrobust  LLR p-value:        0.002939
=====
```

```
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
const  -11.0639   7.231  -1.530   0.126  -25.237   3.109
age      0.2603   0.168   1.549   0.121   -0.069   0.590
=====
```

Odds Ratio:

const 0.000016

age 1.297355

dtype: float64

LogisticRegression()

<matplotlib.collections.PathCollection object at 0x000001C43E44EE90>

[<matplotlib.lines.Line2D object at 0x000001C43E44EFD0>]

Text(0.5, 1.0, 'Age vs Disease')

Accuracy: 0.8

[[4 1]

[1 4]]

[<matplotlib.lines.Line2D object at 0x000001C4438B74D0>]

